



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B.Sc Chemistry				
Course Title	INORGANIC CHEMISTRY-IV				
Type of Course	MAJOR				
Semester	VI				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Solid foundation in Coordination Chemistry. Uniqueness of carbon, covalent bond, and coordinate bond. Bonding in carbon monoxide. Knowledge about minerals and ores. Importance of metal ions in biological systems. Atomic number, mass number, isotopes, basic idea about nuclear radioactivity, Nuclear fission and fusion.				
Course Summary	This course enables the students to develop knowledge about theories of bonding in coordination compounds, stability constants, chelating effects. It covers the detailed study on structure, bonding and applications of organometallic compounds. It gives basic understanding of different metallurgical processes. It enables the students to familiarise with various metal ions, their compounds and their important in biological systems. It covers various aspects of nuclear chemistry including nuclear stability and reactions.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	To Understand and apply theories of bonding in coordination compounds	U	C	Instructor-created exams / Assignments

CO2	To Classify , interpret and create organometallic compounds, including their applications in synthesis and as catalysts	U	C	Instructor-created exams / Assignments
CO3	To Describe and perform steps in metallurgy and recognize the composition and uses of various alloys	Ap	C	Assignment / seminar/quizzes
CO4	To Identify the role of metal ions in biological systems and understand their significance in biological processes	Ap	C	Assignment/Seminar/Class test
CO5	To describe knowledge of the fundamental aspects and practical applications of nuclear chemistry.	Ap	U	Assignment/class test/
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Mark
I	THEORIES OF BONDING IN COORDINATION COMPOUNDS		15	30
	1	Molecular orbital theory-Composition of ligand group orbitals.	2	
	2	MO diagram of octahedral tetrahedral and square planar complexes with and without π -bonding.	4	
	3	Stepwise and overall formation constant and relationship between them, Trends in stepwise formation constants.	3	
	4	Determination of stability constants by spectrophotometric methods. Stabilization of unusual oxidation state.	3	
	5	Thermodynamic origin of chelating effect, Macrocyclic and Template effect.	3	

II	ORGANOMETALLIC CHEMISTRY		12	25
	6	Definition, Classification of organometallic compounds on the basis of M-C bond with examples. Classification of organic ligands based on hapticity.	2	
	7	Metal carbonyls-Definition- Classification, 18 electron rule and deviation from 18 electron rule, Electron count of mononuclear and polynuclear metal carbonyls (calculation by oxidation number method and covalent method). MOT and the basis for the 18 electron rule.	2	
	8	General methods of preparation properties and structures of mono and binuclear carbonyls of Cr, Mn, Fe, Co and Ni.	2	
	8	Bonding in metal carbonyls (MO diagram of CO to be discussed). Synergic effect and use of IR spectra to explain the extent of back bonding.	2	
	9	Preparation, Structure, Bonding and Reactions : Zeise's salt, Metallocenes- Ferrocene (VBT and MOT).	3	
	10	Applications of organometallic compounds in synthesis and as catalysts, Hydrogenation using Wilkinson's catalyst and Polymerization of alkenes using Ziegler Natta catalyst (mechanism not needed).	1	
III	METALLURGY		11	23
	11	Discuss the terms: Mineral, Ore, Gangue, Flux, slag, Electrometallurgy – Hydrometallurgy.	2	
	12	Steps in metallurgy:(a)Pulverization of the ore (b)Concentration of the ore (Physical and chemical) (c)Treatment of the concentrated ore-Calcination and roasting (d)Reduction-different methods: Smelting, Gold smith aluminothermic process, Kroll's process, Electrolytic reduction, Self reduction. (e) Refining :Liquation, Distillation.	2	
	13	Vapour Phase refining, Zone refining, Oxidative refining, Electrolytic refining, Poling, Cupellation, Parting process, Ion exchange method.	2	

	14	Ellingham diagrams for metal oxides – Extractive metallurgy of Fe, Ni and Ti.	3	
	15	Alloys: Definition – Composition and uses of German silver, Brass, Bronze, Gunmetal and Alnico. Steel: Open hearth process, Classification of steel, Composition and uses of alloy steels, Composition, Properties and Applications of industrially important stainless steel types, (AISI) (a brief study).	2	
IV	BIOINORGANIC CHEMISTRY		10	20
	16	Discuss various elements present in the biological system, Essential and Non-essential elements, Metal ions in biological system, Trace and bulk metal ions, Role of alkali metal ions in biological systems, Sodium-potassium pump, Structural role of calcium.	2	
	17	Ligands present in biological systems, Structure of Porphyrin and Corrin.	1	
	18	Structure of heme - Oxygen transport by heme proteins, Hemoglobin and Myoglobin, Structure of the oxygen binding site, Nature of heme- dioxygen binding, Cooperativity.	2	
	19	Structure of Hemerythrin and Hemocyanin.	1	
	20	Metalloenzymes and Metal activated enzymes, Biochemistry of Zn – structure and functions of Carboxypeptidase, Carbonic Anhydrase, Biochemistry of Cobalt, Vitamin B 12 and Deficiency diseases.	2	
	21	Chlorophyll and Photosynthesis (mechanism not expected).	1	
	22	Anticancer drugs. Cis-platin, Oxaliplatin, Carboplatin and Auranofin – Structure and Significance.	1	
V	NUCLEAR CHEMISTRY		12	
	23	The teacher can choose the important topics in the area nuclear chemistry: Nuclear stability, N/P ratio, Mass defect (numerical problems) , Packing Fraction, Binding Energy per Nucleons, Radioactivity, Group displacement law, Disintegration series, Nuclear fission, Fusion, Reactors: Applications, RadioCarbon dating, Rock dating, Isotopes as tracers.	12	

		Must cover Nuclear stability, n/p ratio, Fission, Fusion, Separation of isotopes, Application of isotopes.		
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References:

1. D. F. Shriver, P. W. Atkins, Inorganic Chemistry, 5th Edn., Oxford University Press, New York, 2010.
2. M. C. Day, J. Selbin, Theoretical Inorganic Chemistry, East West Press, New Delhi, 2002.
3. J. E. Huheey, E. A. Keitler, R. L. Keitler, Inorganic Chemistry – Principles of Structure and Reactivity, 4th Edn., Pearson Education, New Delhi, 2013.
4. J. D. Lee, Concise Inorganic Chemistry, 5th Edn., Oxford University Press, New Delhi, 2008.
5. B. R. Puri, L. R. Sharma, K. C. Kalia, Principles of Inorganic Chemistry, 31st Edn., Milestone Publishers and Distributors, New Delhi, 2013.
6. F.A. Cotton, G. Wilkinson, Advanced Inorganic Chemistry, 6TH Edn., Wiley India Pvt. Ltd., New Delhi, 2009.
7. R. Gopalan, V. Ramalingam, Concise Coordination Chemistry, 1st Edn., Vikas Publishing House, New Delhi, 2001.
8. R. C. Mehrotra, A. Singh, Organometallic chemistry, New age publishers, 1991.
9. A. Cottrell, An introduction to metallurgy, 2nd Edn., University press, 1990.
10. Bertini, H. B. Gray, S. J. Lippard, J. Selverstone Valentine, Bioinorganic Chemistry, Viva Books Pvt. Ltd., 2007.
11. P. Powell, Principles of Organometallic Compounds, 2nd Edn., Chapman and Hall, London, 1988.

Mapping of COs with PSOs and POs :

	PS O1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO 1	3	1	-	-	-	1	3				1		1
CO 2	2		-	-	1	2	2		1		2	1	2
CO 3	3	-		-	2	2	3			1			1
CO 4	2				2	1	3					1	1
CO 5	2		-	-	1	3	3					1	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/viva/ seminar	Practical skill evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3		✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		